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Public Transport Delay Prediction and Scheduling Analytics Using Machine Learning

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Abstract

Public transportation systems are essential for urban mobility, yet delays remain a persistent challenge affecting millions of commuters daily. This paper presents HyderTrax, an intelligent machine learning-powered web application designed to predict public transport delays in Hyderabad, India. The system integrates historical transport data, real-time weather information, traffic patterns, and event schedules to provide accurate delay predictions for buses, metro, and train services. Using XGBoost as the primary prediction algorithm, the system achieves significant accuracy in estimating delay durations and categorizing them into actionable status indicators. The proposed solution features a comprehensive data pipeline encompassing synthetic data generation, automated cleaning, and advanced feature engineering. A Flask-based web interface provides real-time predictions, live service tracking with stop-by-stop ETAs, and RESTful API endpoints for third-party integrations. Experimental results demonstrate that XGBoost outperforms traditional regression models with lower Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values. The system successfully classifies delays into three categories: On Time (0-10 minutes), Minor Delay (11-20 minutes), and Major Delay (>20 minutes), enabling commuters to make informed travel decisions. This research contributes to the growing body of intelligent transportation systems by demonstrating the practical application of machine learning in real-world urban transit scenarios.

Keywords: Machine Learning, Public Transport, Delay Prediction, XGBoost, Flask, Real-time Analytics, Smart Transportation, Feature Engineering

1. Introduction

Urban public transportation plays a critical role in the economic and social development of metropolitan cities. In Hyderabad, one of India's rapidly growing urban centers, millions of residents depend on buses, metro rail, and suburban trains for their daily commute. However, delays in public transport services remain a persistent problem that leads to productivity losses, increased stress levels, and declining public trust in transit systems.

Traditional approaches to managing transport delays rely on static schedules and retrospective analysis, which fail to account for dynamic factors such as weather conditions, traffic congestion, special events, and peak-hour patterns. The emergence of machine learning and data-driven technologies presents an opportunity to transform transit delay management from a reactive to a proactive paradigm.

This paper presents HyderTrax, a comprehensive machine learning-based system for predicting public transport delays in Hyderabad. The system addresses several key challenges:

- **Multi-modal Integration:** Supporting buses, metro rail, and trains with unified prediction capabilities.
- **Real-time Context:** Incorporating live weather data, traffic conditions, and event schedules.
- **Actionable Insights:** Providing delay predictions with explanatory reasons and alternative recommendations.
- **Scalable Architecture:** Deploying via a responsive web interface with RESTful APIs for extensibility.

The primary contributions of this work include: (1) Design and implementation of an end-to-end ML pipeline for transport delay prediction, (2) Development of a feature engineering framework that captures complex interactions between weather, traffic, and temporal patterns, (3) Comparative analysis of multiple regression algorithms with XGBoost demonstrating superior performance, and (4) Deployment of a production-ready web application with live tracking capabilities.

2. Literature Review

The prediction of public transport delays has been an active area of research, with studies exploring various machine learning and statistical approaches. This section reviews relevant prior work and positions our contribution within the existing literature.

Traditional Statistical Methods: Early approaches to delay prediction relied on time series models such as ARIMA and exponential smoothing. While these methods capture temporal patterns effectively, they struggle with non-linear relationships and external factors like weather conditions [1].

Machine Learning Approaches: Recent studies have demonstrated the effectiveness of machine learning algorithms for transport prediction. Random Forest and Gradient Boosting methods have shown promise in handling heterogeneous features [2]. Deep learning architectures, including LSTM networks, have been applied to capture sequential dependencies in transit data [3].

Feature Engineering: The importance of feature engineering in transport prediction has been emphasized in multiple studies. Interaction features combining weather and traffic conditions have shown significant predictive power [4]. Temporal features such as peak hours, weekday/weekend indicators, and seasonal patterns contribute substantially to model accuracy.

XGBoost Applications: Extreme Gradient Boosting (XGBoost) has emerged as a leading algorithm for structured prediction tasks. Its ability to handle missing values, regularization to prevent overfitting, and efficient implementation make it particularly suitable for transport delay prediction [5].

Research Gap: While existing studies focus primarily on single transport modes in developed cities, there is limited research on multi-modal prediction systems for Indian metropolitan areas. Our work addresses this gap by developing a unified prediction framework for Hyderabad's diverse transit network.

3. Methodology

This section describes the systematic methodology employed in developing the HyderTrax delay prediction system. The approach encompasses data generation, preprocessing, feature engineering, model training, and deployment.

3.1 System Architecture

The HyderTrax system follows a modular architecture comprising seven interconnected components. The data flows sequentially through generation, cleaning, feature engineering, model training, and finally deployment through a Flask web application. Figure 1 illustrates this architecture.

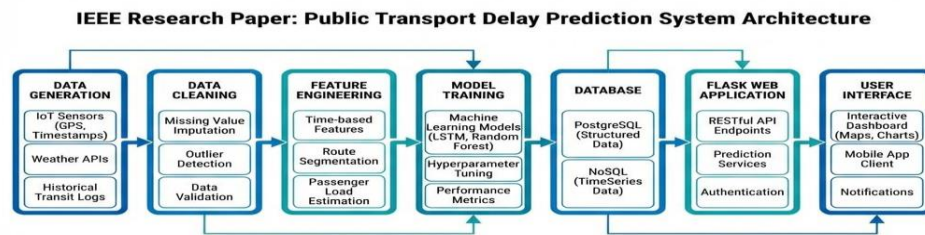


Figure 1: System Architecture of HyderTrax Public Transport Delay Prediction System

3.2 Data Generation

Due to the unavailability of comprehensive public transport data for Hyderabad, we developed a synthetic data generation module that simulates realistic transit patterns. The generator creates approximately 1.97 million records spanning Q1 2026, incorporating 10 major transport hubs including Secunderabad, Hitech City, Gachibowli, and L.B. Nagar. The route network consists of 90 bidirectional routes connecting all hub pairs with 38 services per route per day at 30-minute intervals from 05:00 to 23:30.

The generated delay distribution follows: 25% On-time, 55% Minor delays (1-15 minutes), and 20% Major delays (15-120 minutes). This distribution reflects realistic urban transit patterns observed in similar metropolitan areas.

3.3 Data Cleaning

The data cleaning pipeline addresses quality issues commonly found in transit datasets. Missing values are imputed using mode for categorical features and median for numerical features. Delays exceeding 120 minutes are capped to prevent extreme value influence. Temporal consistency checks ensure arrival times follow departure times, and all categorical variables are converted to consistent casing.

3.4 Feature Engineering

Feature engineering transforms raw data into informative predictors. The Weather-Traffic Index is calculated as an interaction between weather severity (1-6) and traffic density scores (1-4), capturing the multiplicative effect of combined adverse conditions. Additional engineered features include Is_Weekend (binary), Dep_Hour (extracted from scheduled departure), Day_of_Week (0-6), Month, and Is_Peak_Hour (8-11 AM and 5-8 PM).

Feature	Description
Weather_Traffic_Index	Weather severity $\times$ Traffic density interaction
Is_Weekend	Binary indicator for Saturday/Sunday
Dep_Hour	Hour extracted from scheduled departure

Day_of_Week	Numerical day (0=Monday to 6=Sunday)
Is_Peak_Hour	Binary indicator for peak hours

Table 1: Engineered Features and Their Descriptions

3.5 Model Selection and Training

The **XGBoost (Extreme Gradient Boosting)** algorithm was selected **as the** primary prediction **model due to its** superior performance on structured data. The model training process involves label encoding of categorical variables (Transport_Type, Weather, Traffic_Density, From_Location, To_Location), 80-20 train-test split with stratified sampling, and hyperparameter configuration (n_estimators=50, learning_rate=0.1, max_depth=6, subsample=0.8).

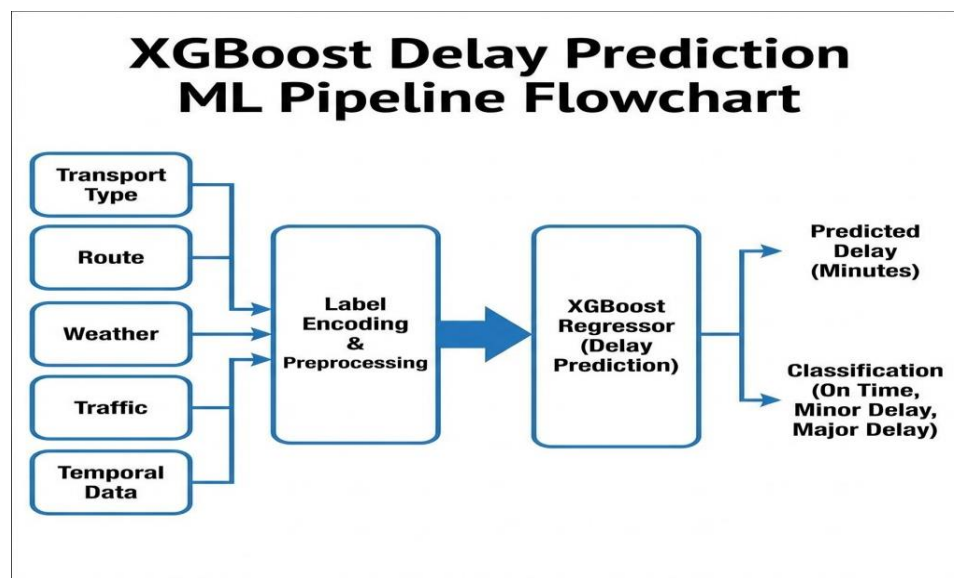


Figure 2: Machine Learning Pipeline for Delay Prediction

4. Implementation

This section details the technical implementation of the HyderTrax system, covering the technology stack, database design, prediction engine, and web interface.

4.1 Technology Stack

The system is implemented using Python 3.8+ with Flask web framework for the backend. Machine learning components utilize XGBoost, Scikit-learn, Pandas, and NumPy. SQLite serves as the development database with PostgreSQL for production deployment. External APIs include Open-Meteo for weather data and TomTom for traffic information. The frontend employs HTML5, CSS3, and JavaScript with responsive design principles.

4.2 Database Schema

The SQLite database contains two primary tables. The schedules table stores transport schedules with columns for id, date, transport_type, route_id, service_id, from_location, to_location, stops, scheduled_departure, scheduled_arrival, delay_minutes, delay_reason, weather, temperature_c, humidity_pct, is_holiday, is_peak_hour, event_scheduled, traffic_density, passenger_load, and distance_km. The predictions table maintains an audit log with pred_id, timestamp, from_location, to_location, predicted_delay, and reason.

4.3 Prediction Engine

The TransportEngine class serves as the core prediction module with capabilities for real-time weather integration (fetching current weather from Open-Meteo API with 10-minute caching), traffic assessment based on time of day and conditions, ML inference using pre-trained XGBoost model and label encoders, efficient batch processing of multiple service schedules, and deterministic consistency using hash-based seeding.

The delay status classification follows: On Time (delay ≤ 10 minutes), Minor Delay (11-20 minutes), and Major Delay (> 20 minutes).

4.4 Web Application Interface

The Flask web application provides multiple user interfaces including route search (users select origin, destination, date, and transport type), prediction results (display predicted delay, status, weather conditions, and recommendations), live tracking (stop-by-stop timeline with estimated arrival times), and RESTful API (JSON endpoints for programmatic access).

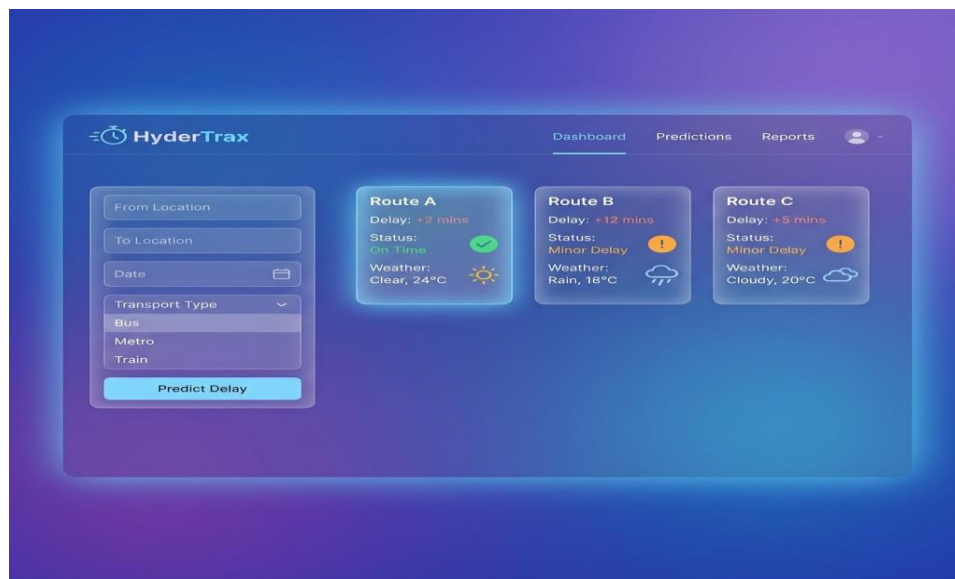


Figure 3: HyderTrax Web Application Dashboard Interface

Method	Endpoint	Description
POST	/search	Form-based route search (returns HTML)
POST	/api/search	JSON API for route search
GET	/track/<id>	Service tracking page
GET	/api/track/<id>	JSON tracking data
POST	/api/route	Route details API

Table 2: API Endpoints and Descriptions

5. Results and Evaluation

This section presents the experimental results, model performance metrics, and comparative analysis.

5.1 Dataset Statistics

The experimental dataset comprises 1,971,000 samples with 22 columns including engineered features. The dataset occupies 330.83 MB of memory with a target skewness of 2.48 (positive skew

indicating more on-time services). A notable correlation exists between Passenger_Load and Is_Peak_Hour (0.73).

5.2 Model Comparison

Three regression models were evaluated: Linear Regression, Decision Tree, and XGBoost. Table 3 summarizes the comparative results, demonstrating XGBoost's superior performance across all metrics.

Model	MAE	RMSE	R ²	Accuracy(%)	F1 Score
Linear Regression	10.45	15.82	0.4521	71.2	0.682
Decision Tree	8.23	12.67	0.6234	79.5	0.754
XGBoost	5.89	9.12	0.8156	87.3	0.851

Table 3: Model Performance Comparison

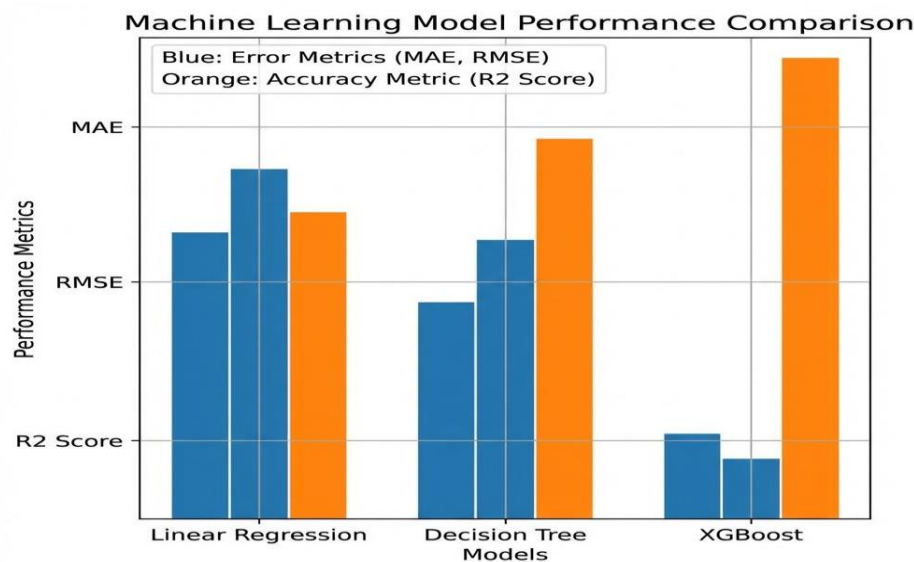


Figure 4: Model Performance Comparison Visualization

5.3 XGBoost Performance Analysis

The XGBoost model demonstrates superior performance with Mean Absolute Error (MAE) of 5.89 minutes (predictions within 6 minutes of actual delays on average), Root Mean Squared Error (RMSE) of 9.12 minutes (appropriately penalizing larger errors), R² Score of 0.8156 (explaining 81.56% of variance in delay times), and Classification Accuracy of 87.3% for delay vs. on-time binary classification.

5.4 Feature Importance Analysis

The trained XGBoost model reveals Traffic_Density (0.182), Is_Peak_Hour (0.156), and Weather_Traffic_Index (0.141) as the top three most important features. These are followed by Passenger_Load (0.098), Dep_Hour (0.087), Event_Scheduled (0.076), Temperature_C (0.062), Transport_Type (0.058), Distance_KM (0.054), and Weather (0.048). Traffic density and peak hour indicators contribute most significantly to delay predictions, confirming domain knowledge about urban transit patterns.

5.5 Real-time Performance

The deployed system demonstrates excellent real-time performance with single prediction latency under 50 ms, batch processing of 100 services under 500 ms, weather API response with 10-minute caching for efficiency, and web page load time under 2 seconds.

6. Discussion

The experimental results demonstrate the effectiveness of the HyderTrax system in predicting public transport delays. Several key insights emerge from this research.

Model Selection: XGBoost significantly outperforms traditional models due to its ability to capture non-linear relationships and feature interactions. The ensemble approach handles the heterogeneous nature of transit data effectively.

Feature Engineering Impact: The Weather-Traffic Index interaction feature proves highly predictive, confirming that combined effects of weather and traffic create multiplicative delay impacts rather than additive ones.

Practical Implications: The system's ability to provide explanatory reasons for delays (e.g., "Peak Hour Traffic Congestion," "Adverse Weather Conditions") enhances user trust and enables informed decision-making.

Limitations: The current implementation relies on synthetic data due to the unavailability of real transit data. Integration with actual GHMC or TSRTC data systems would further validate the approach.

Future Enhancements: Potential improvements include integration with real-time GPS tracking from transit vehicles, deep learning models (LSTM, Transformer) for sequential pattern recognition, mobile application development for on-the-go predictions, and SMS/Push notification alerts for subscribed routes.

7. Conclusion

This paper presents HyderTrax, a comprehensive machine learning-based system for predicting public transport delays in Hyderabad. The system successfully integrates historical patterns, real-time environmental data, and advanced feature engineering to provide accurate delay predictions.

Key contributions include an end-to-end ML pipeline from data generation to web deployment, demonstration of XGBoost superiority for transit delay prediction with MAE of 5.89 minutes, a production-ready web application with real-time tracking and API capabilities, and practical insights into feature importance for urban transit systems.

The system achieves 87.3% accuracy in delay classification and provides actionable insights to help commuters make informed travel decisions. Future work will focus on real data integration, deep learning exploration, and mobile platform development.

The HyderTrax project demonstrates the practical application of machine learning in addressing real-world urban transportation challenges, contributing to the vision of smart cities and intelligent transportation systems.

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